

Suppose u, v, w are vectors of the n -space $\mathbb{R}^n$. Is $\text{span}\{u, v, w\}$ a subspace of $\mathbb{R}^n$?
Is $\{u, v, w\}$ a basis of $\text{span}\{u, v, w\}$? Is the dimension of $\text{span}\{u, v, w\}$ given by n ?

Answer:

Let's explore each part of your question with detailed analysis:

1. Is $\text{span}\{u, v, w\}$ a subspace of $\mathbb{R}^n$?
 - **Yes**, $\text{span}\{u, v, w\}$ is a subspace of $\mathbb{R}^n$. By definition, the span of a set of vectors is the set of all possible linear combinations of those vectors. This set satisfies the conditions for being a subspace:
 - It contains the zero vector (when all coefficients in the linear combination are zero).
 - It is closed under vector addition (adding any two linear combinations results in another linear combination).
 - It is closed under scalar multiplication (scaling a linear combination by any scalar results in another linear combination).
2. Is $\{u, v, w\}$ a basis of $\text{span}\{u, v, w\}$?
 - **Not necessarily.** A set of vectors $\{u, v, w\}$ is a basis for $\text{span}\{u, v, w\}$ only if the set is linearly independent. To determine if $\{u, v, w\}$ forms a basis:
 - The vectors must be linearly independent, meaning that no vector in the set can be written as a linear combination of the others.
 - If they are linearly independent, then $\{u, v, w\}$ is indeed a basis for $\text{span}\{u, v, w\}$.
 - If they are not linearly independent, such as one being a linear combination of others, then there are redundant vectors, and the set is not a basis.
3. The dimension of $\text{span}\{u, v, w\}$ depends on the number of linearly independent vectors. It is at most 3 if u, v, w are all linearly independent.
 - **No**, the dimension of $\text{span}\{u, v, w\}$ is not necessarily n . The dimension of the span is equal to the number of vectors in its basis, which is the maximum number of linearly independent vectors that can be extracted from $\{u, v, w\}$.
 - The dimension equals n only if $\text{span}\{u, v, w\}$ completely fills $\mathbb{R}^n$ which would require at least n linearly independent vectors, so this typically isn't the case unless $n \leq 3$.

These concepts are consistent with materials found in course documents and common vector space theory. Each vector's ability to act as a basis element is contingent on the absence of redundancy, ensuring independence among them.